Analysis of Factorial Experiment Using R

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Loading libraries

```
library(readxl)
library(ExpDes)
library(agricolae)
library(tidyverse)
library(ggstatsplot)
library(car)
library(patchwork)
library(emmeans)
library(multcomp)
```

Native tree experiment

- Factor A: spacing (10x10 m2, 12x12 m2)
- Factor B: age at planting (6, 12, 24 mo.)
- Response: height (ft)

Import and explore data

```
#Import the excel file into R
fact_crd <- read_excel("Factorial.xlsx",
                       sheet = "CRD")

#Display the first 6 rows of the data frame
head(fact_crd)
```

```
## # A tibble: 6 × 4
##       A     B   Rep Height
##   <dbl> <dbl> <dbl>  <dbl>
## 1     1     1     1   46.5
## 2     1     1     2   55.9
## 3     1     1     3   78.7
## 4     1     2     1   49.5
## 5     1     2     2   59.5
## 6     1     2     3   78.7
```

```
with(fact_crd,fat2.crd(factor1 = A,
                       factor2 = B,
                       resp = Height,
                       quali = c(TRUE,TRUE),
                       mcomp = "tukey",
                       fac.names = c("Spacing", "Age")))
```

```
## ------------------------------------------------------------------------
## Legend:
## FACTOR 1:  Spacing
## FACTOR 2:  Age
## ------------------------------------------------------------------------
##
##
## Analysis of Variance Table
## ------------------------------------------------------------------------
##             DF      SS MS      Fc    Pr>Fc
## Spacing      1   409.0  3  1.5207 0.241120
## Age          2 12846.3  5 23.8835 0.000066
## Spacing*Age  2   996.6  4  1.8529 0.198942
## Residuals   12  3227.2  2
## Total       17 17479.1  1
## ------------------------------------------------------------------------
## CV = 20.36 %
##
## ------------------------------------------------------------------------
## Shapiro-Wilk normality test
## p-value:  0.4529232
## According to Shapiro-Wilk normality test at 5% of significance, residuals can
be considered normal.
## ------------------------------------------------------------------------
```

```
##
## No significant interaction: analyzing the simple effect
## ------------------------------------------------------------------------
## Spacing
## According to the F test, the means of this factor are statistical equal.
## ------------------------------------------------------------------------
##   Levels    Means
## 1      1 85.30000
## 2      2 75.76667
## ------------------------------------------------------------------------
## Age
## Tukey's test
## ------------------------------------------------------------------------
## Groups Treatments Means
## a     3   118.0833
##  b    2   65.36667
##  b    1   58.15
## ------------------------------------------------------------------------
```

Graphical display

```
#Create factors from integer codes
fact_crd <- fact_crd %>%
  mutate(Spacing = factor(A, labels = c("10x10", "12x12")),
         Age = factor(B, labels = c("6","12","24")))

head(fact_crd)
```

```
## # A tibble: 6 × 6
##       A     B   Rep Height Spacing Age
##   <dbl> <dbl> <dbl>  <dbl> <fct>   <fct>
## 1     1     1     1   46.5 10x10   6
## 2     1     1     2   55.9 10x10   6
## 3     1     1     3   78.7 10x10   6
## 4     1     2     1   49.5 10x10   12
## 5     1     2     2   59.5 10x10   12
## 6     1     2     3   78.7 10x10   12
```

```r
#Create the ANOVA model
fact_mod1 <- aov(Height ~ Spacing + Age + Spacing:Age,
                 data = fact_crd)

#Display the ANOVA table
summary(fact_mod1)
```

```
##             Df Sum Sq Mean Sq F value   Pr(>F)
## Spacing      1    409     409   1.521    0.241
## Age          2  12846    6423  23.883 6.55e-05 ***
## Spacing:Age  2    997     498   1.853    0.199
## Residuals   12   3227     269
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Graphical display

```
#Generate the summary statistics for the levels of Factor B (Age)
fact_crd <- fact_crd  %>%
  group_by(Age)  %>%
  dplyr::summarize(Mean = mean(Height),
            SD = sd(Height),
            n = length(Height))  %>%
  dplyr::mutate(SE = SD/sqrt(n))

fact_crd
```

```
## # A tibble: 3 × 5
##   Age    Mean    SD     n    SE
##   <fct> <dbl> <dbl> <int> <dbl>
## 1 6      58.2  12.0     6  4.89
## 2 12     65.4  10.4     6  4.24
## 3 24    118.   26.0     6 10.6
```

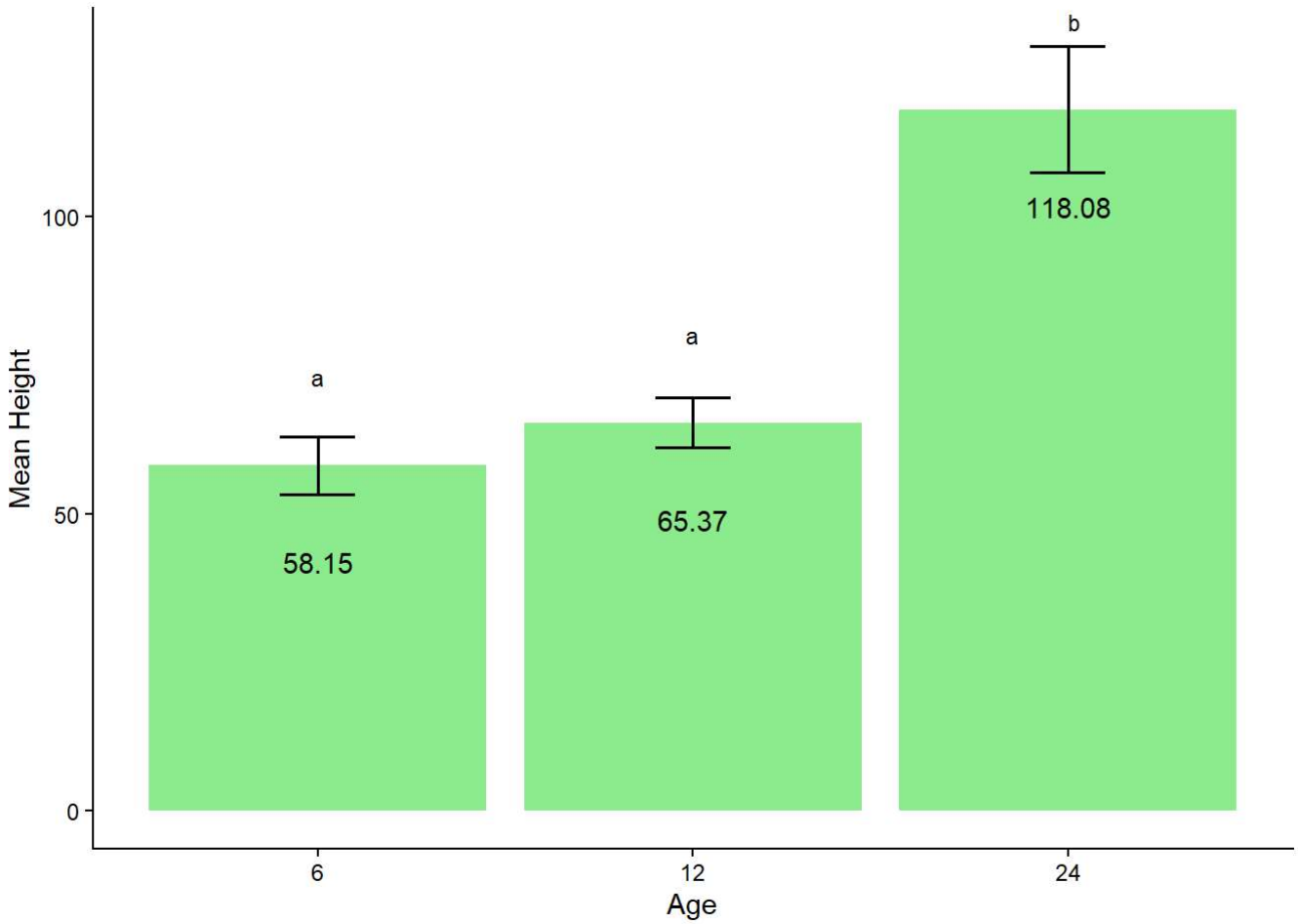
```
B.means <- emmeans(fact_mod1, specs="Age")
B.means
```

```
##  Age emmean   SE df lower.CL upper.CL
##  6     58.1 6.69 12     43.6     72.7
##  12    65.4 6.69 12     50.8     80.0
##  24   118.1 6.69 12    103.5    132.7
##
## Results are averaged over the levels of: Spacing
## Confidence level used: 0.95
```

```
#Assign letters
B.means.cld <- cld(B.means,
                   Letters=letters)  %>%
  arrange(Age)  %>%
  mutate(SE1 = fact_crd$SE)
B.means.cld
```

```
##   Age    emmean       SE df  lower.CL  upper.CL .group       SE1
## 1   6  58.15000 6.694975 12  43.56290  72.73710     a   4.888882
## 2  12  65.36667 6.694975 12  50.77957  79.95376     a   4.237269
## 3  24 118.08333 6.694975 12 103.49624 132.67043      b 10.610008
```

```
#Generate the two-way plot
ggplot(B.means.cld, aes(Age, emmean,label = .group)) +
  geom_col(fill="lightgreen") +
  geom_errorbar(aes(ymin=emmean - SE1, ymax = emmean + SE1),
                width = 0.2, size = 0.7) +
  geom_text(vjust=-4.5, size = 3) +
   geom_text(label=round(B.means.cld$emmean,2),vjust=5, color = "black") +
  labs(x= "Age", y= "Mean Height") +
  theme_classic()
```



Analysis of Factorial Experiments in RCBD

- A factorial experiment was carried out in order to determine the response of one variety of a crop to three levels of a new plant growth regulator (PGR), denoted by P1, P2, P3, and five levels of nitrogen fertilizer (N1, N2, N3, N4, N5)
- A randomized block design with three replications was adopted and the 15 treatments were randomly assigned to the plots within each block

- One of the parameters that were recorded is plot yield (kg/ha)

Importing the data into R

```
fact_rbd <- read_excel("Factorial.xlsx",
                       sheet = "RCBD")
head(fact_rbd)
```

```
## # A tibble: 6 × 4
##   Block     P     N Yield
##   <dbl> <dbl> <dbl> <dbl>
## 1     1     1     1   0.9
## 2     1     1     2   1.2
## 3     1     1     3   1.3
## 4     1     1     4   1.8
## 5     1     1     5   1.1
## 6     1     2     1   0.9
```

Factorial ANOVA (RCBD)

```
with(fact_rbd,fat2.rbd(factor1 = P,
                       factor2 = N,
                       block = Block,
                       resp = Yield,
                       quali = c(TRUE,TRUE),
                       fac.names = c("P","N"),
                       mcomp="tukey"))
```

```
## ------------------------------------------------------------------------
## Legend:
## FACTOR 1:  P
## FACTOR 2:  N
## ------------------------------------------------------------------------
##
##
## Analysis of Variance Table
## ------------------------------------------------------------------------
##           DF     SS MS     Fc    Pr>Fc
## Block      2 0.0640  3  1.631 0.213772
## P          2 0.1693  4  4.316 0.023244
## N          4 2.4902  6 31.732 0.000000
## P*N        8 1.0151  5  6.468 0.000090
## Residuals 28 0.5493  2
## Total     44 4.2880  1
## ------------------------------------------------------------------------
## CV = 11.12 %
##
## ------------------------------------------------------------------------
## Shapiro-Wilk normality test
## p-value:  0.1377132
## According to Shapiro-Wilk normality test at 5% of significance, residuals can
be considered normal.
```

```
## ------------------------------------------------------------------------
##
##
##
## Significant interaction: analyzing the interaction
## ------------------------------------------------------------------------
##
## Analyzing  P  inside of each level of  N
## ------------------------------------------------------------------------
## ------------------------------------------------------------------------
## Analysis of Variance Table
## ------------------------------------------------------------------------
##           DF      SS      MS      Fc  Pr.Fc
## Block      2 0.06400   0.032  1.6311 0.2138
## N          4 2.49022 0.62256 31.7322      0
## P:N 1      2 0.01556 0.00778  0.3964 0.6764
## P:N 2      2 0.12667 0.06333  3.2282 0.0548
## P:N 3      2 0.01556 0.00778  0.3964 0.6764
## P:N 4      2 0.62000    0.31  15.801      0
## P:N 5      2 0.40667 0.20333 10.3641  4e-04
## Residuals 28 0.54933 0.01962
## Total     44 4.28800
## ------------------------------------------------------------------------
##
##
```

```
##
##  P  inside of the level  1  of  N
##
## According to the F test, the means of this factor are statistical equal.
## ----------------------------------------------------------------------
##     Levels     Means
## 1        1 0.9333333
## 2        2 0.8333333
## 3        3 0.8666667
## ----------------------------------------------------------------------
##
##
##  P  inside of the level  2  of  N
##
## According to the F test, the means of this factor are statistical equal.
## ----------------------------------------------------------------------
##     Levels     Means
## 1        1 1.2333333
## 2        2 0.9666667
## 3        3 1.2000000
## ----------------------------------------------------------------------
##
##
##  P  inside of the level  3  of  N
##
```

```
## According to the F test, the means of this factor are statistical equal.
## ------------------------------------------------------------------------
##     Levels     Means
## 1        1  1.400000
## 2        2  1.300000
## 3        3  1.366667
## ------------------------------------------------------------------------
##
##
##  P  inside of the level  4  of  N
## ------------------------------------------------------------------------
## Tukey's test
## ------------------------------------------------------------------------
## Groups Treatments Means
## a     1   1.933333
##  b    3   1.433333
##  b    2   1.333333
## ------------------------------------------------------------------------
##
##
##  P  inside of the level  5  of  N
## ------------------------------------------------------------------------
## Tukey's test
## ------------------------------------------------------------------------
## Groups Treatments Means
```

```
## a     2   1.666667
##  b    1   1.233333
##  b    3   1.2
## ----------------------------------------------------------------------
##
##
##
## Analyzing  N  inside of each level of  P
## ----------------------------------------------------------------------
## ----------------------------------------------------------------------
## Analysis of Variance Table
## ----------------------------------------------------------------------
##           DF      SS      MS      Fc  Pr.Fc
## Block      2 0.06400   0.032  1.6311 0.2138
## P          2 0.16933 0.08467  4.3155 0.0232
## N:P 1      4 1.63067 0.40767 20.7791      0
## N:P 2      4 1.29733 0.32433 16.5316      0
## N:P 3      4 0.57733 0.14433  7.3568  4e-04
## Residuals 28 0.54933 0.01962
## Total     44 4.28800
## ----------------------------------------------------------------------
##
##
##
##  N  inside of the level  1  of  P
```

```
## ------------------------------------------------------------------------
## Tukey's test
## ------------------------------------------------------------------------
## Groups Treatments Means
## a     4   1.933333
##  b    3   1.4
##  bc   2   1.233333
##  bc   5   1.233333
##   c   1   0.9333333
## ------------------------------------------------------------------------
##
##
##  N  inside of the level  2  of  P
## ------------------------------------------------------------------------
## Tukey's test
## ------------------------------------------------------------------------
## Groups Treatments Means
## a     5   1.666667
##  b    4   1.333333
##  b    3   1.3
##   c   2   0.9666667
##   c   1   0.8333333
## ------------------------------------------------------------------------
##
##
```

```
##  N  inside of the level  3  of  P
## ------------------------------------------------------------------------
## Tukey's test
## ------------------------------------------------------------------------
## Groups Treatments Means
## a     4   1.433333
## a     3   1.366667
## a     2   1.2
## a     5   1.2
##  b    1   0.8666667
## ------------------------------------------------------------------------
```

Graphical display

```
#Converting the integer codes to factors
fact_rbd <- fact_rbd %>%
  mutate(Block = factor(Block),
         PGR = factor(P, labels = c("P1", "P2", "P3")),
         Nitro = factor(N, labels = c("N1", "N2", "N3", "N4", "N5")))
head(fact_rbd)
```

```
## # A tibble: 6 × 6
##   Block     P     N Yield PGR   Nitro
##   <fct> <dbl> <dbl> <dbl> <fct> <fct>
## 1 1         1     1   0.9 P1    N1
## 2 1         1     2   1.2 P1    N2
## 3 1         1     3   1.3 P1    N3
## 4 1         1     4   1.8 P1    N4
## 5 1         1     5   1.1 P1    N5
## 6 1         2     1   0.9 P2    N1
```

```
#Create the ANOVA model
fact_mod2 <- aov(Yield ~ Block + PGR + Nitro + PGR:Nitro,
                 data = fact_rbd)

#Display the ANOVA table
anova(fact_mod2)
```

```
## Analysis of Variance Table
##
## Response: Yield
##           Df  Sum Sq Mean Sq F value    Pr(>F)
## Block      2 0.06400 0.03200  1.6311   0.21377
## PGR        2 0.16933 0.08467  4.3155   0.02324 *
## Nitro      4 2.49022 0.62256 31.7322 4.946e-10 ***
## PGR:Nitro  8 1.01511 0.12689  6.4676 8.979e-05 ***
## Residuals 28 0.54933 0.01962
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#Generate summary statistics for all treatment combinations
pn.means <- fact_rbd  %>%
  group_by(P,N)  %>%
  dplyr::summarise(Mean = mean(Yield),
            SD = sd(Yield),
            n = length(Yield))  %>%
  dplyr::mutate(SE = SD/sqrt(n))

pn.means
```

```
## # A tibble: 15 × 6
## # Groups:   P [3]
##        P     N  Mean     SD     n     SE
##    <dbl> <dbl> <dbl>  <dbl> <int>  <dbl>
##  1     1     1 0.933 0.0577     3 0.0333
##  2     1     2 1.23  0.0577     3 0.0333
##  3     1     3 1.4   0.1        3 0.0577
##  4     1     4 1.93  0.153      3 0.0882
##  5     1     5 1.23  0.153      3 0.0882
##  6     2     1 0.833 0.0577     3 0.0333
##  7     2     2 0.967 0.115      3 0.0667
##  8     2     3 1.3   0.2        3 0.115
##  9     2     4 1.33  0.252      3 0.145
## 10     2     5 1.67  0.208      3 0.120
## 11     3     1 0.867 0.153      3 0.0882
## 12     3     2 1.2   0.2        3 0.115
## 13     3     3 1.37  0.0577     3 0.0333
## 14     3     4 1.43  0.0577     3 0.0333
## 15     3     5 1.2   0.1        3 0.0577
```

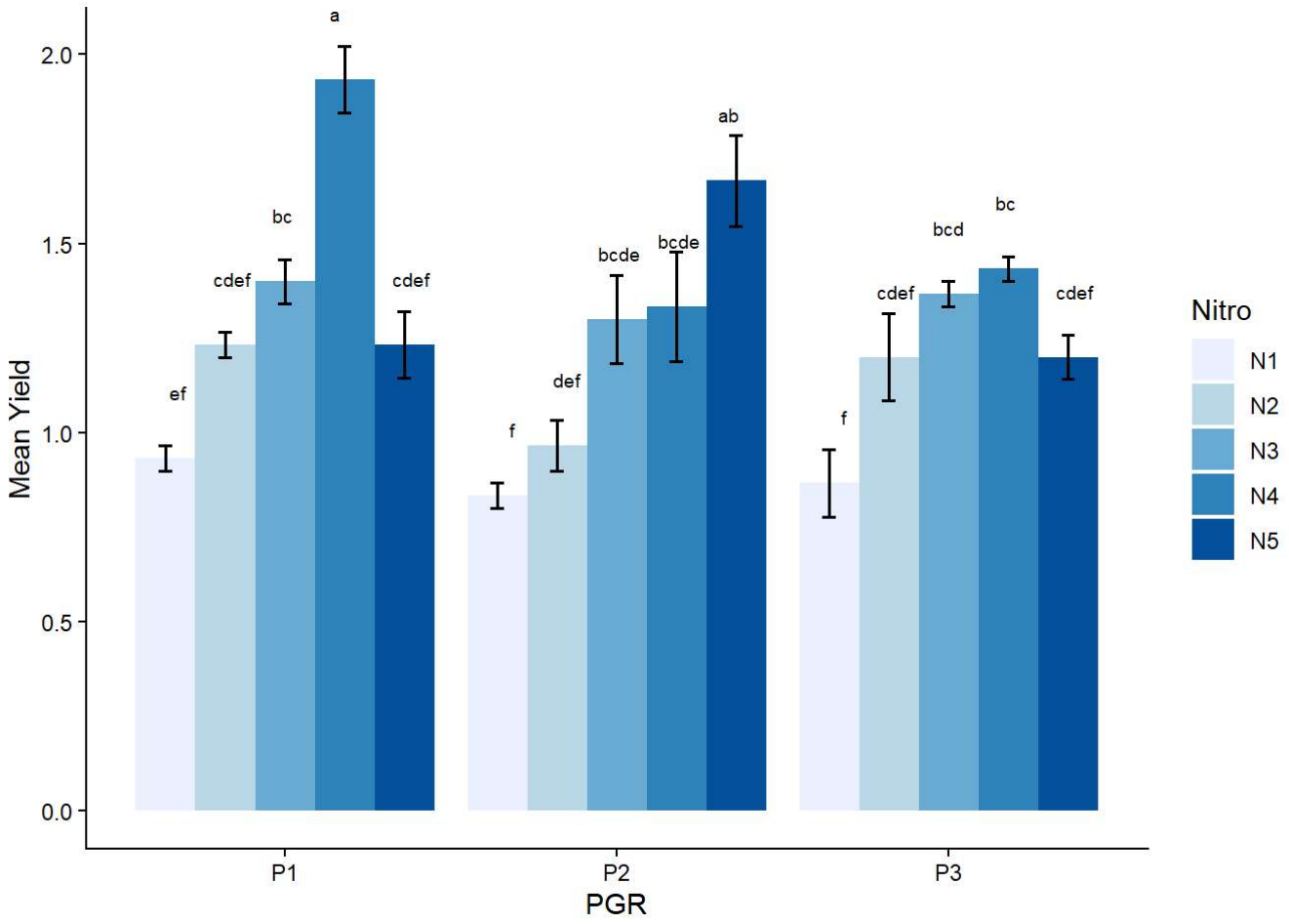
```
PN.means <- emmeans(fact_mod2, specs=c("PGR", "Nitro"))
PN.means
```

```
##  PGR Nitro emmean     SE df lower.CL upper.CL
##  P1  N1     0.933 0.0809 28    0.768    1.099
##  P2  N1     0.833 0.0809 28    0.668    0.999
##  P3  N1     0.867 0.0809 28    0.701    1.032
##  P1  N2     1.233 0.0809 28    1.068    1.399
##  P2  N2     0.967 0.0809 28    0.801    1.132
##  P3  N2     1.200 0.0809 28    1.034    1.366
##  P1  N3     1.400 0.0809 28    1.234    1.566
##  P2  N3     1.300 0.0809 28    1.134    1.466
##  P3  N3     1.367 0.0809 28    1.201    1.532
##  P1  N4     1.933 0.0809 28    1.768    2.099
##  P2  N4     1.333 0.0809 28    1.168    1.499
##  P3  N4     1.433 0.0809 28    1.268    1.599
##  P1  N5     1.233 0.0809 28    1.068    1.399
##  P2  N5     1.667 0.0809 28    1.501    1.832
##  P3  N5     1.200 0.0809 28    1.034    1.366
##
## Results are averaged over the levels of: Block
## Confidence level used: 0.95
```

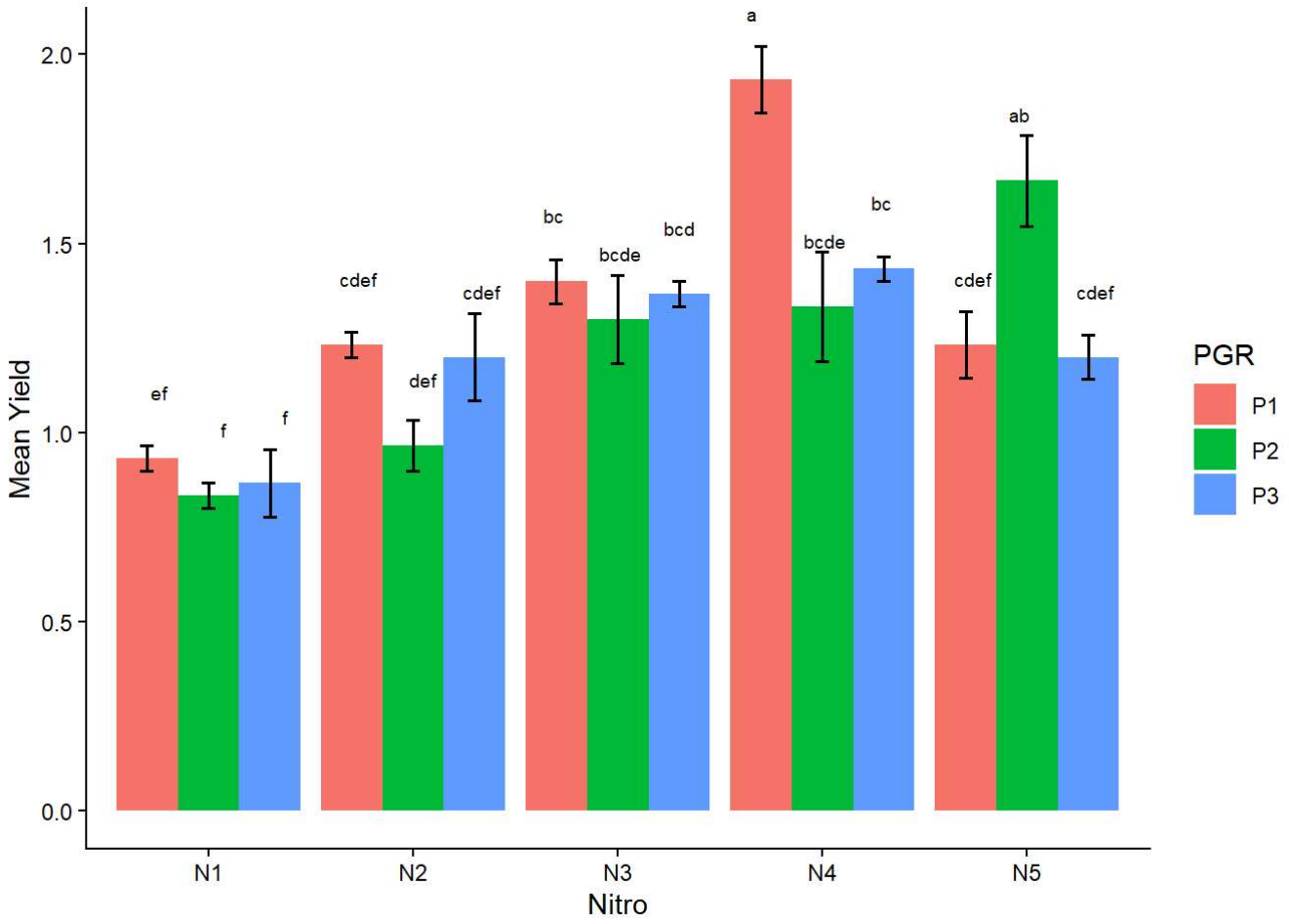
```
#AGenerate the letters
PN.means.cld <- cld(PN.means,
                   Letters=letters,
                   decreasing = TRUE)

PN.means.cld <- PN.means.cld  %>%
  dplyr::arrange(PGR,Nitro)  %>%
  dplyr::mutate(SE1 = pn.means$SE)

PN.means.cld
```

```
##    PGR Nitro    emmean        SE df  lower.CL  upper.CL  .group        SE1
## 1   P1    N1 0.9333333 0.0808683 28 0.7676821 1.0989845      ef 0.03333333
## 2   P1    N2 1.2333333 0.0808683 28 1.0676821 1.3989845    cdef 0.03333333
## 3   P1    N3 1.4000000 0.0808683 28 1.2343488 1.5656512   bc    0.05773503
## 4   P1    N4 1.9333333 0.0808683 28 1.7676821 2.0989845  a      0.08819171
## 5   P1    N5 1.2333333 0.0808683 28 1.0676821 1.3989845    cdef 0.08819171
## 6   P2    N1 0.8333333 0.0808683 28 0.6676821 0.9989845       f 0.03333333
## 7   P2    N2 0.9666667 0.0808683 28 0.8010155 1.1323179     def 0.06666667
## 8   P2    N3 1.3000000 0.0808683 28 1.1343488 1.4656512   bcde  0.11547005
## 9   P2    N4 1.3333333 0.0808683 28 1.1676821 1.4989845   bcde  0.14529663
## 10  P2    N5 1.6666667 0.0808683 28 1.5010155 1.8323179  ab     0.12018504
## 11  P3    N1 0.8666667 0.0808683 28 0.7010155 1.0323179       f 0.08819171
## 12  P3    N2 1.2000000 0.0808683 28 1.0343488 1.3656512    cdef 0.11547005
## 13  P3    N3 1.3666667 0.0808683 28 1.2010155 1.5323179   bcd   0.03333333
## 14  P3    N4 1.4333333 0.0808683 28 1.2676821 1.5989845   bc    0.03333333
## 15  P3    N5 1.2000000 0.0808683 28 1.0343488 1.3656512    cdef 0.05773503
```

```
#Comparing the levels of N for each level of P
ggplot(PN.means.cld, aes(x = PGR,
                         y = emmean,
                         fill = Nitro,
                         label = .group)) +
  geom_col(position = "dodge") +
  geom_errorbar(aes(ymin = emmean - SE1,
                    ymax = emmean + SE1),
                width = 0.2,
                size = 0.7,
                position = position_dodge(0.9)) +
  geom_text(vjust=-4,
            position = position_dodge(0.9),
            size = 2.5) +
  labs(x= "PGR", y= "Mean Yield") +
  scale_fill_brewer(palette = "Blues") +
  theme_classic()
```


Mean Yield
2.0
1.5
1.0
0.5
0.0
a
ab
bc
bcde
bcde
bcd
bc
cdef
cdef
cdef
cdef
cdef
ef
def
f
f
P1
P2
P3
PGR
Nitro
N1
N2
N3
N4
N5

```
#Comparing the levels of P for each level of N
ggplot(PN.means.cld, aes(x = Nitro,
                         y = emmean,
                         fill = PGR,
                         label = .group)) +
  geom_col(position = "dodge") +
  geom_errorbar(aes(ymin = emmean - SE1,
                    ymax = emmean + SE1),
                width = 0.2,
                size = 0.7,
                position = position_dodge(0.9)) +
  geom_text(vjust=-4,
            position = position_dodge(0.9),
            size = 2.5) +
  labs(x= "Nitro", y= "Mean Yield") +
  theme_classic()
```


Mean Yield
2.0
1.5
1.0
0.5
0.0
ef
f
f
cdef
def
cdef
bc
bcde
bcd
a
bcde
bc
cdef
ab
cdef
N1
N2
N3
N4
N5
Nitro
PGR
P1
P2
P3